# A Lexicalist Analysis towards Resultative Verb Constructions in Benue-Kwa Languages

Jakob Maché
jakob.mache@letras.ulisboa.pt

Faculdade de Letras
Centro de Linguística da Universidade de Lisboa
Universidede de Lisboa

*50 anos do CLUP* Universidade do Porto
9th November 2026



Centro de Linguística
da Universidade de Lisboa

CLUL

Overview

Introduction
Properties, other adjectival concepts and stative verbs in Benue-Kwa

Tense and Aspect in Benue-Kwa

Resultative Serial Verb Constructions in Benue-Kwa
- Syntax
- Aspectual properties

Analysis
- Previous work
- A lexicalist approach
- Aspectual Semantics in Lexicalist Frameworks
- Type Theory with Records
- Ingredients
- Lexicon rule
- Lexicon rule
- Lexicon entry for complex predicate $V_1$

Conclusions

CLUL

# Handout




https://dkaramasov.github.io/mache/assets/pdf/26-a-lexicalist-analysis-towards-resultative-verb-constructions-CLUP-HO.pdf

CLUL

# Special thanks to

- Robin Cooper
- Jonathan Ginzburg
- Comlan Atanase Degbevi
- Victor Manfredi
- Olúwàdára Ọmọtọ̀ṣọ́
- Chinedu Úchèchúkwu
- Abídémi Jimoh
- Waliyat Adérojú
- Stefan Müller
- John Beavers
- Antonio Machicao y Priemer
- Ìbùkúnolú Ajagùnnà
- Hana Filip

CLUL

# Resultative predicates in Germanic languages

Simpson (1983, 153 Fn.2), Boas (2003, pp. 8–11), S. M. Wechsler (2001, p. 4), S. Wechsler (2005, pp. 256–257), Müller (2002, pp. 210–211), Goldberg and Jackendoff (2004, p. 536), Beavers (2012, p. 909): Germanic resultative predicates are complex predicates, where the means predicate is a V and the result predicate is syntactically realised as AdjP or PP – but never as VP:

(1) Gaylord pulled the door [$_{AP}$ open].

(2) Gaylord pulled the door [$_{PP}$ from the truck].

CLUL

# Resultative Serial Verb Constructions in Benue-Kwa

Lord (1974, 1975, pp. 24–28), Baker (1989, pp. 529–535), Déchaine (1993, p. 807), Collins (1997, pp. 472–477), Stewart (2001, pp. 149–151): Resultative Serial Verb Constructions (RSVC) consist of two V/(verbal) roots/VPs:

(3) Ó$_i$ pọ́n ___$_i$ rà.[1]
it ripen rot

'It ripened to the point of rotting.' YORÙBÁ

(4) Àsíbá$_i$ dɔ̀n [ví ɔ́]$_j$ [$_{VP}$___$_j$ yì kɔ́xɔ̀].[2]
Àsíbá pull child DET go out.side

'Àsíbá pulled the child outside.' FɔN

- Assumption here: RSVC as resultative predicates, where the result predicate is a V
- Alternative view: result predicate in Ẹ̀dó e.g. *dé* 'fall' as AdjP or PP (cf. Stewart 2001, pp. 28–29, 37)

[1] As quoted in Oyèlárăn (1982, p. 120) = ex. (19b).
[2] As quoted in Lefebvre and Brousseau (2002, p. 423) =ex. (66b).

CLUL

# Consequences and challenges I

- Two verbs can belong to different aspectual classes and overall RSVC
  - *fà ṣí* ‘pull open’ (YORÙBÁ ): telic and gradable ACCOMPLISHMENT
  - *fà* ‘pull’: atelic ACTIVITY
  - *ṣí* ‘open’: telic and gradable ACCOMPLISHMENT/atelic STATE,
- Question how does the event composition function in detail?
- Are there any rules to determine the aspectual class of the overall RSVC?

CLUL

# Consequences and challenges II

- Syntactic properties are determined by $V_1$ ($V_1$ is head)
- Aspectual properties are determined/?influenced by $VP_2$

CLUL

# The data

- Existing data from literature
- New data from Yorùbá and Fɔn
  - former foreign language students born between 1990 and 2000
  - collected between January 2021 till March 2023 on site, via email and chats
  - elicited through translation tasks
  - cross checked through rating tasks

CLUL

# Inchoative verbs, stative verbs and adjectival concepts in Benue-Kwa I

Many Benue-Kwa languages express properties and ‘adjectival concepts’ primarily as stative verbs in Yorùbá (cf. Awóbùlúyì 1967, pp. 180–182, Madugu 1976), Ìgbo (cf. Nwachukwu 1984, pp. 83–84, Nwachukwu 1987, pp. 19–20), Fɔn (cf. Lefebvre and Brousseau 2002, pp. 348–349) or Efutu (Guang) (cf. Agyeman 2016, pp. 135–142):

1. They undergo the same kind of nominalisation via partial C*í*CV-reduplication (cf. Madugu 1976, pp. 94–97)

(5) [$_{NP}$Gíga ọmọ náà] wù mí[3]
NMLZ-be.tall child the please me

‘I like the child’s tallness.’ YORÙBÁ

[3] As quoted in Madugu (1976, p. 90) = ex. (16).

# Inchoative verbs, stative verbs and adjectival concepts in Benue-Kwa II

2. They take the same kind of preverbal TAM-modifiers (Madugu 1976, pp. 88–91 for Yorùbá , HABITUAL *na* in Ewe can precede a verb cf. Ameka 2008, pp. 137–139)

(6) Ọmọ náà ń ga.[4]
child DET PROG tall
'The child is getting tall.' YORÙBÁ

(7) Ọmọ náà ti ga.[5]
child DET PRF tall
'The child has become tall.' YORÙBÁ

[4] As quoted in Madugu (1976, p. 90) = ex. (12).
[5] As quoted in Madugu (1976, p. 90) = ex. (12).

# Inchoative verbs, stative verbs and adjectival concepts in Benue-Kwa III

3. They participate in comparative serial verb constructions

   (8) Ọmọ náà ga ju Olu (lọ)[6]
   child the be.tall than Olu SURPASS
   'The child is taller than Olu.'

4. Yorùbá: require HIGH TONE SYLLABLE just as verb do, as well in Arigidi (Defoid) (cf. Oshòdì 2011, pp. 264, 267) and Àbèsàbèsì (Akpe, cf. Lau 2021, p. 72)
5. Ìgbo : Some of these properties require the *r*V-suffix (cf. Nwachukwu 1984, pp. 83–84)

[6] As quoted in Madugu (1976, p. 90) = ex. (17).

# Inchoative verbs, stative verbs and adjectival concepts in Benue-Kwa IV

Most of these languages small(er) set of ‘closed class’ more marked items: e.g. 6 for Ìgbo (cf. Nwachukwu 1984, pp. 97–99), 18 for Fɔn (cf.Lefebvre and Brousseau 2002, pp. 350–352):

1. They occur adnominal as attributes (cf. Nwachukwu 1984, pp. 97–99 for Ìgbo , Lefebvre and Brousseau 2002, pp. 350–352 for Fɔn , Agyeman 2016, pp. 137–138 for Efutu)

(9) [$_{NP}$nwókó ọ́cha.]
man PFX-be.light.colored?

‘a fair complexioned man’ ÌGBO

# ‘Genuine’ adjectival concepts in Benue-Kwa II

2. Do not participate in C*í*CV-reduplication, as they are reduplicated forms or nominalisations already (cf. Madugu 1976, pp. 94–97)

(10) [$_{NP}$ọmọ kékéré/dúdú.]
child RED-small/black

‘a small/black child’ YORÙBÁ

3. Lefebvre and Brousseau (2002, p. 351) ‘genuine’ adjectives cannot occur as the main predicate in Fɔn , but require a copula, similar in Efutu (cf. Agyeman 2016, pp. 137, 171–172)

(11) a. * Kɔ̀kú ɖàgbè / ɖàgbó / vú / glì.
Kɔ̀kú good / important / young / short
Intended: ‘Kɔ̀kú is good/important/young/short.’
FɔN

b. Kɔ̀kú ɖò ɖàgbè / ɖàgbó / vú / glì.
Kɔ̀kú be.at good / important / young / short

# Verbs and gradability

In some languages verbs can express gradable states and events: they take degree intensifiers or occur in *wie*-exclamatives

(12) Sie hat sich sehr/extrem gefreut.
she has REFL much/extremely be.happy
'She was very/extremely happy.'

(13) Wie (sehr) der Helmerl heute wieder auszuckt.
how much DET Helmerl today again get.mad
'How much Helmerl got upset again today!'

CLUL

# Tense and Aspect in Benue-Kwa

| | BK1 (old) Àkán , Ẹ̀dó , Ìgbo , Bantu, . . . | BK2 (new) Gbè, Yorùbá , . . . |
|---|---|---|
| PAST/PRESENT ambiguity | — | ✓ |
| only one event per clause | — | ✓ |
| pro-clitic inflection | — | ✓ |
| three surface tones | — | ✓ |

Figure: Two large groups of languages among Benue-Kwa according to Manfredi (2005, p. 2)

CLUL

# The interpretation of unmarked/tenseless verbs in BK 2: The FACTATIVE

Benue-Kwa 2 languages make use of unmarked finite verb forms such, as Yorùbá (cf. Awóbùlúyì 1967, pp. 260–261, 265, Welmers 1973, pp. 346–347 and Awóyalé 1991, p. 218) and Gungbe (cf. Aboh and Nauze 2008, pp. 217–218) with factative interpretation:

- Default interpretation of eventive predicates: past tense
- Default interpretation of stative predicates: present tense
- Some telic ACHIEVEMENT/ACCOMPLISHMENT verbs are ambiguous between a PRESENT and a PRESENT PERFECT interpretation (cf. Avolonto 1992, pp. 97–98 and Lefebvre and Brousseau 2002, pp. 86–89 for Fɔn )

Extends to Western Kwa: Efutu (Guang), as illustrated by Agyeman (2016, pp. 186–190)

CLUL

# The interpretation of unmarked/tenseless verbs in BK 2: Alternative view

Essegbey (1999, pp. 32, 64–67) and Ameka (2008, pp. 135–136, 139–140) argue for Ewe-Gbè: Only two types of verb classes:

1. “Action verbs” (≈ ACTIVITIES)
2. Inchoatives (≈ ACHIEVMENTS and ACCOMPLISHMENTS): Stative verbs can be systematically derived from inchoatives

Plausible analysis for Yorùbá inchoatives and statives (cf. Manfredi pers. commun)

# Inchoative and stative interpretations

(14) a. Òkè náà ga.[7]
mountain DET tall

‘The mountain is high.’ YORÙBÁ

b. Òkè náà ń ga.
mountain DET PROG tall

‘The mountain is becoming high.’ YORÙBÁ

#‘The mountain is being high.’

- Two possible conclusions:
  1. ‘Stative’ predicates can be coerced into inchoative interpretations
  2. Or: inchoative is the base form which only becomes visible in certain contexts
- Assumption here: Option 2; unmarked verbs have default interpretations, but stative interpretations always derived from eventive ones

[7] Elicited examples (translation task).

# The interpretation of tensed verbs in BK 1: Ìgbo

All eventive and most stative predicates in Ìgbo are inflected with the so-called (FACTATIVE) *r*V-suffix (cf. Welmers 1973, pp. 346–347 and Manfredi 1991, p. 174).

(15) a. * Ógu tóro.[8]
Ógu tall-rV
Intended: ‘Ógu is/became tall. ÌGBO

b. Ógu tóro eto
Ógu tall-rV NMLZ-tall
‘Ógu is/became tall. ÌGBO

- Default interpretation with most stative verbs: present tense.
- Default interpretation with eventive verbs: past tense.
- Inchoatives mostly derive a stative interpretation
- The *r*V-suffix can never be clause final: intransitive verbs have BOUND VERB COMPLEMENT

[8] As quoted in Nwachukwu (1987, pp. 19–20) = ex. (12a–b).

# The interpretation of tensed verbs in BK 1: Ẹ̀dó

Manfredi (2022, p. 5): Also Ẹ̀dó predicates can remain ambiguos between a inchoative/telic interpretation and a stative interpretation in the presence of the (FACTATIVE) (*r*)-ẹ̀ suffix:

(16) a. Ọ̀ gọ́-(r)ẹ̀.[9]
3S.S bend-FACT

‘It became bent.’ ẸDÓ

‘It is [now] bent.’

b. Ọ̀ guọghọ́-(r)ẹ̀.[10]
3S.S break-FACT

‘It broke.’ ẸDÓ

‘It is [now] broken.’

[9] As quoted in Manfredi (2022, p. 5) = ex. (25a).
[10] As quoted in Manfredi (2022, p. 5) = ex. (25b).

# Resultative Serial Verb Constructions – Yorùbá

*jẹ* ⟨AG, TH⟩ ‘eat’ + *tán* ⟨TH⟩ ‘be.finish’ → *jẹ* X *tán* ⟨AG, TH⟩ ‘eat X be.finish’

(17) Ẹniọlá á jẹ [ọbẹ̀ ewédú] tán.
Ẹniọlá HTS eat soup jute.leaf be.finished
‘Ẹniọlá finished the ewédú.’ (SVC)

a. Ẹniọlá á jẹ [ọbẹ̀ ewédú].
Ẹniọlá HTS eat soup jute.leaf
‘Ẹniọlá ate ewédú.’ (TRANS)

b. Ọbẹ̀ ewédú ú tán.
soup jute.leaf HTS be.finished
‘The ewédú soup is finished.’ (UNACC)

- HIGH TONE SYLLABLE precedes $V_1$ and mostly limited to veridical contexts

CLUL

# Syntactic headedness

- Imperative possible with full RSVC

(18) Jẹ ọbẹ ewédú tán!
eat soup jute.leaf be.finished
'Finish up the ewédú!' YORÙBÁ

- Imperative possible with $V_1$

(19) Jẹ ọbẹ ewédú!
eat soup jute.leaf
'Eat the ewédú!' YORÙBÁ

- Sometimes imperative impossible with $V_2$

(20) # tán!
be.finished
Intended:'Be finished!' YORÙBÁ

- $V_1$ is the syntactic head, similar conclusion by Ogie (2009, pp. 476–479)

CLUL

# Possible base for derivational morphology

- Event nominalisation derived from RSVC-bases: C*í*+(CV)$_V$

(21) ọbẹ̀ewédú jí[$_V$jẹ-tán]!
soup-jute.leaf NMLZ-eat-be.finished
'the jute leaf soup up eating' YORÙBÁ

- Role nouns derived from RSVC-bases: *a*- + V

(22) a-j=ọbẹ̀ewédú-tán!
NMLZ-eat=soup-jute.leaf-be.finished
' the ewédú soup up eater' YORÙBÁ

- Calls for a lexical analysis for RSVC

CLUL

# Syntactic microvariation within Benue-Kwa I

Largely absent from Gbè-Cluster, Ka-Togo and Na-Togo, as illustrated for FɔN -Gbè

(23) Kɔ̀kú$_i$ xò [kòklô ɔ́]$_j$ __$_{i/*j}$ hù.[11]
Kɔ̀kú hit chicken DEF kill
‘ Kɔ̀kú beat the chicken to death.’ FɔN

(24) * Kɔ̀kú$_i$ xò [kòklô ɔ́]$_j$ __$_j$ kú.[12]
Kɔ̀kú hit chicken DEF die

[11] As quoted in Lefebvre and Brousseau (2002, p. 414) = ex. (41).
[12] As quoted in Lefebvre and Brousseau (2002, p. 415) = ex. (42).

# Syntactic microvariation within Benue-Kwa II

- Same for other Gbè-languages: εuε Ameka (2003, pp. 9–12; 2005, pp. 361–364, 373), Gun (cf. Aboh 2009, pp. 21–23)
- Similar data from Ka-Togo: Avatime (cf. Defina 2016, p. 100), Tutrugbu (cf. Essegbey 2019, pp. 245–246), Tuwuli (cf. Harley 2005, pp. 432–451)
- Na-Togo: Likpe (cf. Ameka 2003, p. 11)
- Outside Benue-Ka: Dàgáárè (Gur, Hiraiwa and Bodomo 2008, pp. 808–809)

CLUL

# Aspectual properties of resultative predicates in Germanic

- S. M. Wechsler (2001, p. 4, 2005, p. 258): Canonical resultative predicates are always telic
- S. M. Wechsler (2001, pp. 9–12, 2005, pp. 264–268) and Beavers, p. 925 (2012, p. 925, 2013, pp. 690–692) two large subtypes of resultative predicates
  1. ACCOMPLISHMENTS: durative V + gradable closed-scale adjective, e.g. *Mary hammered the metal flat*
  2. ACHIEVEMENTS: punctual V + non-gradable, closed-scale maximum endpoint Adj, e.g. *He shot the miller dead*

CLUL

# Modeling telicity according to Beavers (2013, pp. 684–688)

(25) a. John warmed the pie to 100° in/?for an hour.

b. $\exists e \exists s[\textit{warming}'(\mathbf{j}, \mathbf{p}, s, e) \wedge \textit{result}'(s, \mathbf{100°}, e)]$

(26) a. John warmed the pie for/??in an hour.

b. $\exists e \exists g \exists s[\textit{warming}'(\mathbf{j}, \mathbf{p}, s, e) \wedge \textit{result}'(s, g, e)]$

- Telic predicates (quantised, cf. 25) come with specified goal arguments *g* and a simple/complex scale *s*.
- Some atelic predicates (non-quantised, cf. 26) have unspecified goal arguments *g*
- There is a relation which defines *g* as the result of the event *e*: result(s,g,e)

CLUL

# Aspectual classes of RSVC in Benue-Kwa

Ogie (2009, pp. 235–238): Benue-Kwa also has two types of RSVC: gradable and durative ACCOMPLISHMENTS (cf. 27) and punctual non-gradable ACHIEVEMENTS (28)

(27) Òzó$_i$ họ̀ọ́ úkpọ̀n$_j$ __$_j$ huán.[13]
Òzó wash.PST cloth be.clean.PST
'Òzó washed the clothes clean.' ỊDÓ

(28) Òzó suá Àzàrí [$_{VP}$dé gbé òtò] vbé
Òzó push.PST Àzàrí fall.PST against ground at
ágógó èvá/ *ífuánrò évá.[14]
clock two minute two
'Òzó pushed Àzàrí down at 2pm/*in two minutes.'

[13] As quoted in Ogie (2009, p. 236) =ex. (69a).
[14] As quoted in Ogie (2009, p. 237) =ex. (70b).

# Atelic readings with RSVC

Challenge: Durative adverbs may coerce atelic readings with RSVC: goal will be interpreted as unspecified on the content value of the modifier.

(29) Olú ti ọmọ náà ṣubú fún wákàtí kan.[15]
Olú push child DET fall give hour one
'Olú pushed the child for one hour' (ITERATIVE)
'Olú pushed the child for one hour' (CONTINUOUS)

(30) Fẹ́mi í fọ̀ aṣọ náà mọ́n fún wákàti kan.[16]
Fẹ́mi HTS wash cloth DET clean for hour one
'Fẹ́mi washed the cloth clean/thoroughly for an hour.'
(CONTINUOUS)

[15] Elicited example (translation task).
[16] Elicited example (translation task).

# Previous work: Derivational Approaches

Analysis of interpretation of the whole RSVC: Stewart (2001, p. 27), Zimmermann and Amaechi (2025)

(31) a. Òzó sùá ọ̀gọ́ dé.[17]
Òzó push bottle fall
'Òzó pushed the bottle down.' ẸDÓ

b. (∃*e*) [Push-Fall(e) & AGENT(e, Òzó) & THEME(e,ọ̀gọ́)]

(32) a. Ó gbù-dà-rà ósísí.[18]
3S.S chop-fall-rV tree
'He felled the tree.' ÌGBO

b. ∃*e*∃*s* [AG(Maria,e) ∧ chop(e) ∧ TH(tree,e) ∧ Caus(e,s) fallen(s) ∧ HOLD(tree,s)]

[17] As quoted in Stewart (2001, p. 27) = ex. (14a–b).
[18] As quoted in Zimmermann and Amaechi (2025, p. 45) = ex. (93a–b).

# Previous work: Type-Logical Grammar

Naumann and Gamerschlag (2021, pp. 354–355) provide a constructor rule which built a RSVC from two simple predicates (33) and an example of the type *push* X *fall* (34).

(33) $\lambda P_1.\lambda P_2.\lambda x.\lambda y.\exists e_1.\exists e_2[P_1(e_1) \wedge P_2(e_2) \wedge \textit{arg-pattern}(e_1, e_2), x, y) \wedge \textit{relation}(e, e_1, e_2)]$[19]

(34) $\lambda x.\lambda y.\lambda e.\exists e_1.\exists e_2[\text{push}(e_1) \wedge \text{fall}(e_2) \wedge \text{actor}(e_1) = x \wedge \text{theme}(e_1) = y = \text{theme}(e_2) \wedge e = e_1\ e_2 \wedge \textit{cause}(e_1, e_2)\ ]$
[20]

A further analysis within HPSG/Minimal Recursion Semantic for Ẹ̀dó is provided by Ogie (2009, pp. 477–480), mostly occupied with argument identification.

[19] As quoted in Naumann and Gamerschlag (2021, p. 354) = ex. (35).
[20] As quoted in Naumann and Gamerschlag (2021, p. 355) = ex. (37).

# RSVC in Benue-Kwa II

# Motivation

- RSVC can act as base for derivational morphology
  - Role nouns derived from RSVC-bases: *a-* + V
  - C*í*+(CV)$_V$ event nominalisation
- General assumption: verbal nominalisation applies to V-heads, rather than full fledged sentences

CLUL

# Ingredients for the analysis for object-oriented RSVC

1. Lexicon rules transforming $V_{trans}$ $\langle NP_{AG}, NP_{TH} \rangle$ into $V_{trans\text{-}res}$ selecting for $\langle NP_{AG}, NP_{TH}, Pred \rangle$
   - event composition: *macroevent e* is a String Type composed of concatenation of events expressed by the verbal components (cf. Bohnemeyer, Enfield, et al. 2007, pp. 497, 502–508 and Bohnemeyer and Van Valin 2017, pp. 144–148, Fernando 2004, . . . )
   - goal argument *g* inherited from the result predicate $VP_2$: contextually given *g* necessary for a telic interpretation
   - scale $\zeta$ inherited the result predicate $VP_2$: determines the type of scale: simplex/complex
   - Following Beavers (2013, pp. 684–688), there is the a relation which defines *g* as the result of the event *e*: result($\zeta$,g,$e_2$)
2. phrasal immediate dominance schemes combining complex predicate with predicate argument

CLUL

# Event Structure Templates

Van Valin and Lapolla (1997, pp. 102–113) and Rappaport Hovav and Levin (1998, pp. 109–111): UG provides small number of universal event structure templates

(35) a. ACTIVITIES: [$x$ ACT$_{\langle MANNER \rangle}$ ($y$)]

b. STATES: [$x$ $\langle STATE \rangle$]

c. ACHIEVEMENTS: [BECOME [$x$ $\langle$STATE$\rangle$]]

d. ACCOMPLISHMENTS: [$x$ CAUSE [BECOME [z $\langle$STATE$\rangle$]]]

e. ACCOMPLISHMENTS: [[$x$ ACT$_{\langle \text{MANNER} \rangle}$ ($y$) ] CAUSE [BECOME [$z$ $\langle$STATE$\rangle$]]]

- Rappaport Hovav and Levin (1998, pp. 97, 123, 127–130) projected into syntax from the lexicon entry of the V
- based on four aspectual classes
- cover all verb meanings

CLUL

# Template Augmentation rules

This can be ensured by means of a formalisation of *Template Augmentation Rule* inspired by Rappaport Hovav and Levin (1998, pp. 118–123)

- Lexical rule which takes a transitive ACTIVITY $V_1$ as input and returns a resultative transitive ACCOMPLISHMENT $V_1$
- Adds an additional intransitive ACHIEVEMENT $V_1$ verb on its ARG-ST list of $V_1$
  turns ACHIEVEMENT predicates in possible result states, which lack the MEP and therefore qualify as $V_2$ in a RSVC
- This $V_2$ has a defective event structure lacking the macro event property
- Complex ACCOMPLISHMENT $V_1$ only denotes a single complex event

CLUL

Cooper (2005, p. 99), Cooper (2023) attempt to fuse several theories into one formalism:

1. Montague Semantics ($\lambda$-calculus)
2. Situation Semantics (following tradition of Jon Barwise)
3. Discourse Representation Theory
4. HPSG
5. Probablistic semantics
6. Incremental processing/Dynamic Syntax
7. Semantic frames

Analysis here implemented in *Conversation Oriented Semantics* (*KoS*) as proposed by Ginzburg (2012)

Situation Semantics: Situation type corresponds to the set of possible worlds consistent with the state affairs described by the situation: eg. worlds in which someone hugs someone (cf. 36a), or a boy hugs a dog (cf. 36a):

(36) a.
$$\begin{bmatrix} x & : & \text{Ind} \\ y & : & \text{Ind} \\ e & : & \text{hug(x,y)} \end{bmatrix}$$

b.
$$\begin{bmatrix} x & : & \text{Ind} \\ c_1 & : & \text{boy(x)} \\ y & : & \text{Ind} \\ c_2 & : & \text{dog(y)} \\ e & : & \text{hug(x,y)} \end{bmatrix}$$

c. $\exists x\,[\text{boy}(x) \wedge \exists y\,[\text{dog}(y) \wedge \text{hug}(x,y)]]$

CLUL

TTR assumes a rich ontology of types, which are represented as Record Types.

(37) a. $\langle e, \langle e, t \rangle \rangle$

b. $\lambda x \lambda y . [hug'(x, y)]$

c. $\lambda r_1{:}[x^c{:}\mathrm{IND}].\ \lambda r_2{:}[x^s{:}\mathrm{IND}].$

$$\left[\ c_1 \ : \ \mathrm{hug}(r_2.x^s, r_1.x^c)\ \right]$$

d. $\lambda r_1{:}[x^c{:}\mathrm{IND}].\ \lambda r_2{:}[x^s{:}\mathrm{IND}].$

$$\left[\begin{array}{lcl} r_2.x^s & : & \mathrm{IND} \\ r_1.x^c & : & \mathrm{IND} \\ c_1 & : & \mathrm{hug}(r_2.x^s, r_1.x^c) \end{array}\right]$$

# Ingredients

- Lexicon entries with rich event structures
- Lexicon rule that turns transitive ACTIVITY $V_1$ into ACCOMPLISHMENT $V_2$, adding extra defective resultative ACHIEVEMENT predicate $V_2$ to ARGUMENT STRUCTURE list
- Lexicon rule that turns common achievement predicate into $V_2$
- $V_2$ is slightly grammaticalised lacking MEP

CLUL

# Event structure: ACTIVITY

$$\lambda r_1{:}[x^c{:}\mathit{Ind}].\ \lambda r_2{:}[x^s{:}\mathit{Ind}].\ \begin{bmatrix} r_2.x^s{:}\mathit{Ind} \\ r_1.x^c{:}\mathit{Ind} \\ e{=}e\text{-}str.e{:}\mathit{Event} \\ e\text{-}str: \begin{bmatrix} x{:}\mathit{Ind} \\ y{:}\mathit{Ind} \\ e_1{:}\mathit{Event} \\ e{=}e_1{:}\mathit{Event} \\ \zeta{=}\ \neg\mathrm{Boolean}{:}\mathit{Scale} \\ c_{MSTATE}{:}\mathrm{drink}(x,y,\zeta,e_1) \\ c_{ACT}{:}\mathrm{do}(x,c_{STATE},e_1) \end{bmatrix} \end{bmatrix}$$

- More specified notation of $\lambda y.\lambda x.\mathrm{drink}(x, y)$
- First line specifies the types of argument: $2\times$ *Ind* ($e$)
- Second line specifies the $+$MEP
- Third line describes event structure

CLUL

# Event structure: ACCOMPLISHMENT

$$\lambda r_1{:}[x^c{:}\mathit{Ind}].\ \lambda r_2{:}[x^s{:}\mathit{Ind}].\ \begin{bmatrix} r_2.x^s{:}\mathit{Ind} \\ r_1.x^c{:}\mathit{Ind} \\ e{=}e\text{-}str.e{:}\mathit{Occur} \\ e\text{-}str{:} \begin{bmatrix} x{:}\mathit{Ind} \\ y{:}\mathit{Ind} \\ e_1{:}\mathit{Event} \\ e_2{:}\mathit{Event} \\ e{=}e_1 \frown e_2{:}\mathit{Occur} \\ \zeta{=}\neg\mathrm{Boolean}{:}\mathit{Scale} \\ c_{\mathrm{MSTATE}}{:}\mathrm{drink}(x,y,\zeta,e_1) \\ c_{\mathrm{ACT}}{:}\mathrm{do}(x,c_{\mathrm{STATE}},e_1) \\ c_{\mathrm{BECOME}}{:}\mathrm{become}(c_{\mathrm{RSTATE}},\zeta) \\ c_{\mathrm{RSTATE}}{:}\neg\mathrm{exist}(x,y,\zeta,e_2) \end{bmatrix} \end{bmatrix}$$

- More specified notation of $\lambda y.\lambda x.\mathrm{drink.up}(x, y)$
- First line specifies the types of argument: $2\times$ *Ind* ($e$)
- Second line specifies the $+$MEP
- Third line describes event structure

CLUL

# Lexical rule for *Template Augmentation*: Function

- Takes an transitive ACTIVITY V as input
- Returns a transitive ACCOMPLISHMENT V as output
- Adds a defective resultative predicate to argument list
- Resultative predicate is an intransitive ACHIEVEMENT
- Resultative predicate is defective lacks MEP
- Output event structure is merger of empty ACCOMPLISHMENT template with the ACTIVITY $V_1$ and the ACHIEVEMENT $V_2$

CLUL

# Lexical rule for *Template Augmentation*

$$\textbf{vrsvc}: \left[\begin{array}{l} \text{arg-st}=\boxed{1}\oplus\boxed{2}\oplus\langle\, v_2{:}\left[\begin{array}{l}\text{cat:}\left[\begin{array}{l}\text{head=V:}\mathit{PoS}\\ \text{subj=}\boxed{2}\text{:}\mathit{PoS}\end{array}\right]\\ \text{arg-st=}\langle\boxed{2}\rangle\ \text{:}\mathit{list(synsem)}\end{array}\right]\rangle\text{:list(synsem)} \\ \text{cont=}\lambda r_1\text{:}\left[\begin{array}{l}\text{e=e-str.}e_2\text{:}\mathit{Event}\\ \text{e-str=}\boxed{4}\text{:}\left[\begin{array}{l}\text{y=}\boxed{2}\text{:}\mathit{Ind}\\ e_2\text{:}\mathit{Event}\\ \zeta\text{=}\neg\text{Boolean:}\mathit{Scale}\\ c_{\text{BECOME}}\text{:become}(c_{\text{RSTATE}},\zeta)\\ c_{\text{RSTATE}}\text{:stpredicate}(\text{y},\zeta,e_2)\end{array}\right]\end{array}\right].\ \lambda r_2{:}[\text{y}^c\text{=}\boxed{2}\text{:}\mathit{Ind}].\ \lambda r_3{:}[\text{x}^s\text{=}\boxed{1}\text{:}\mathit{Ind}]. \\ \left[\begin{array}{l}\text{e=e-str.e:}\mathit{Occur}\\ \text{e-str:Accomplishment}\dot{\wedge}\boxed{3}\ \dot{\wedge}\boxed{4}\end{array}\right]\text{:}\mathit{DTV} \end{array}\right]$$

input-dtr

$$\textbf{tv-dtr}: \left[\begin{array}{l}\text{arg-st=}\boxed{1}\langle \text{np}_1{:}\left[\text{x}^s\text{=cont.x}^s\text{:}\mathit{Ind}\right]\rangle\oplus\boxed{2}\langle \text{np}_2{:}\left[\text{y}^c\text{=cont.y}^c\text{:}\mathit{Ind}\right]\rangle\ \text{:}\mathit{list(synsem)}\\ \text{cont=}\lambda r_2{:}[\text{y}^c\text{=}\boxed{2}\text{:}\mathit{Ind}].\lambda r_3{:}[\text{x}^s\text{=}\boxed{1}\text{:}\mathit{Ind}].\\ \qquad\left[\begin{array}{l}\text{x}^s\text{=}\boxed{1}\text{:}\mathit{Ind}\\ \text{y}^c\text{=}\boxed{2}\text{:}\mathit{Ind}\\ \text{e=e-str.}e_1\text{:}\mathit{Event}\\ \text{e-str=}\boxed{3}\text{:}\mathit{Activity}\end{array}\right]\text{:}\mathit{TV}\end{array}\right]$$

# Relabelling rule for defective ACHIEVEMENT predicate: Function

- Takes an intransitive ACHIEVEMENT V with MEP
- Returns a defective ACHIEVEMENT without MEP
- Relabels subject argument *x* as object argument *y*

# Relabelling rule for defective ACHIEVEMENT predicate

$$\textbf{intr-obj-rel}: \left[\begin{array}{l} \text{cont}= \lambda r_1{:}[\text{y}^s{:}\mathit{Ind}]. \left[\begin{array}{l} r_1.\text{y}^s{=}\text{e-str.y}{:}\mathit{Ind} \\ \text{e}{:}\mathit{Event} \\ \text{e-str}: \left[\begin{array}{l} \text{y}{=}\text{intr-dtr.cont.e-str.x}{:}\mathit{Ind} \\ \text{e}_2{=}\text{intr-dtr.cont.e-str.e}_1{:}\mathit{Event} \\ \zeta{=}\text{intr-dtr.cont.e-str.}\zeta{=}\neg\text{Boolean}{:}\mathit{Scale} \\ \text{c}_{\text{BECOME}}{:}\text{become}(\text{c}_{\text{RSTATE}},\zeta) \\ \text{c}_{\text{RSTATE}}{:}\neg\text{exist}(\text{x},\text{y},\zeta,\text{e}_2) \end{array}\right] \end{array}\right] {:}\mathit{IV} \\ \qquad\qquad \text{input-dtr} \\ \qquad\qquad \Big| \\ \textbf{intr-dtr}: \left[\begin{array}{l} \text{cont}{=}\lambda r_1{:}[\text{x}^s{:}\mathit{Ind}]. \left[\begin{array}{l} r_1.\text{x}^s{=}\text{e-str.x}{:}\mathit{Ind} \\ \text{e}{=}\text{e-str.e}{:}\mathit{Event} \\ \text{e-str}: \left[\begin{array}{l} \text{x}{:}\mathit{Ind} \\ \text{e}_1{:}\mathit{Event} \\ \text{e}{=}\text{e}_1{:}\mathit{Event} \\ \zeta{=}\neg\text{Boolean}{:}\mathit{Scale} \\ \text{c}_{\text{BECOME}}{:}\text{become}(\text{c}_{\text{RSTATE}},\zeta) \\ \text{c}_{\text{RSTATE}}{:}\neg\text{exist}(\text{x},\text{y},\zeta,\text{e}_1) \end{array}\right] \end{array}\right] {:}\mathit{IV} \end{array}\right]$$

Figure: Relabelling Rule for object-oriented intransitive ACHIEVEMENT predicates

CLUL

# Event structure: ACHIEVEMENT

$$\lambda r_1{:}[x^s{:}\mathit{Ind}].\begin{bmatrix} r_1.x^s{=}x.\text{e-str}{:}\mathit{Ind} \\ e{=}\text{e-str}.e{:}\mathit{Event} \\ \text{e-str}{:}\begin{bmatrix} x{:}\mathit{Ind} \\ e_1{:}\mathit{Event} \\ e{=}e_1{:}\mathit{Event} \\ \zeta{=}\neg\text{Boolean}{:}\mathit{Scale} \\ c_{\text{BECOME}}{:}\text{become}(c_{\text{RSTATE}},\zeta) \\ c_{\text{RSTATE}}{:}\neg\text{exist}(x,y,\zeta,e_1) \end{bmatrix} \end{bmatrix}$$

- More specified notation of $\lambda x.\text{finish}(x)$
- First line specifies the type of argument: *Ind* ($e$)
- Second line specifies the $+$MEP
- Third line describes event structure

CLUL

# Conclusions

- Benue-Kwa RSVC are resultative predicates with a V as result predicate
- RSVC can serve as bases for derivational morphology
- RSVC can be built through lexicon rule (Template Augmentation)
  - Which transforms a transitive ACTIVITY $V_1$ into a a ACCOMPLISHMENT
  - Adds a defective ACHIEVEMENT $V_2$ to the ARGUMENT STRUCTURE LIST of $V_1$ (cf. Müller (2002))
  - Defective ACHIEVEMENT $V_2$ lacks MACRO-EVENT PROPERTY
- DefectiveACHIEVEMENT $V_2$ can be generated through another relabelling lexicon rule
- Variation: Gbè, Ka-Togo, Na-Tog languages lack relabelling rule: no RSVC available

CLUL


Aboh, Enoch Oladé (2009). “Clause structure and verb series”. In: *Linguistic Inquiry* 40.1, pp. 1–33. DOI: 10.1162/ling.2009.40.1.1.


Aboh, Enoch Oladé and Fabrice Nauze (2008). “Tense, mood, aspect in Gungbe (Kwa)”. In: *Modality-Aspect Interfaces*. Ed. by Werner Abraham and Elisabeth Leiss. Typological Studies in Language 79. Amsterdam: John Benjamins, pp. 215–239.


Agyeman, Nana Ama (2016). “A descriptive grammar of Efutu (southern Ghana) with a focus on serial verb constructions: a language documentation study.”. PhD thesis. London: School of Oriental and African Studies, University of London. URL: http://eprints.soas.ac.uk/id/eprint/23586.


Ameka, Felix K. (2003). “Multiverb constructions in a West African areal typological perspective”. In: *Proceedings of TROSS03 – Trondheim Summer School: Multi-verb-constructions*.


CLUL

Ed. by Dorothee Beermann and Lars Hellan. Trondheim: Institutt for språk- og kommunikasjonsstudier, Norges teknisk-naturvitenskapelige universitet.


Ameka, Felix K. (2005). “Forms of secondary predication in serializing languages”. In: *Secondary Predication and Adverbial Modification*. Ed. by Nikolaus P. Himmelmann and Eva Schultze-Berndt. Oxford: Oxford University Press, pp. 355–378.


— (2008). “Aspect and Modality in Ewe: a survey”. In: *Aspect and Modality in Kwa Languages*. Ed. by Felix K. Ameka and Mary Esther Kropp Dakubu. Amsterdam: John Benjamins Publishing Company, pp. 135–194.


Avolonto, Aimé (1992). “AspP et la catégorie INF en Fɔ̀ngbè”. In: *The Journal of West African Languages* 22.1, pp. 98–113.


Awóbùlúyì, Ọladélé (1967). “Studies in the syntax of Standard Yoruba”.

CLUL

PhD thesis. New York: Columbia University.

 Awóyalé, Yíwọlá (1991). “The tense system of yoruba”. In: *AAP* 27, pp. 194–224.

 Baker, Mark C. (1989). “Object sharing and projection in serial verb constructions”. In: *Linguistic Inquiry* 20.4, pp. 513–553. URL: http://www.jstor.org/stable/4178644.

 Beavers, John (2012). “Resultative Constructions”. In: *The Oxford Handbook of Tense and Aspect*. Ed. by Robert I. Binnick. Berlin: Oxford University Press, pp. 908–934. DOI: 10.1093/oxfordhb/9780195381979.013.0032.

 — (2013). “Aspectual classes and scales of change”. In: *Linguistics* 51.4, pp. 681–706. DOI: 10.1515/ling-2013-0024.

 Boas, Hans C. (2003). *A constructional approach to resulatives.* Stanford, CA: Center for the Study of Language and Information.

 Bohnemeyer, Jürgen, Nicholas J. Enfield, et al. (2007). “Principles of event

segmentation in language. The case of motion events". In: *Language* 83.3, pp. 495–532. DOI: 10.1111/j.1749-818X.2009.00128.x.


Bohnemeyer, Jürgen and Robert D. Van Valin (2017). "The macro-event property and the layered structure of the clause". In: *Studies in Language* 41.1, pp. 142–197. DOI: 10.1075/sl.41.1.05van.


Collins, Chris Thad (1997). "Argument Sharing in Serial Verb Constructions". In: *Linguistic Inquiry* 28.3, pp. 461–497. URL: https://www.jstor.org/stable/4178987.


Cooper, Robin (2005). "Records and record types in semantic theory". In: *Journal of Logic and Computation* 15.2, pp. 99–112.


— (2023). *From perception to communication: A type theory for action and meaning*. Oxford Studies in Semantics and Pragmatics 16. Oxford University Press. DOI: 10.1093/oso/9780192871312.001.0001.

CLUL

Déchaine, Rose-Marie A. (1993). “Serial verb constructions”. In: *Syntax: an international handbook*. Ed. by Joachim Jacobs. Berlin: De Gruyter, pp. 789–825.

Defina, Rebecca (2016). “Events in Language and Thought: The Case of Serial Verb Constructions in Avatime”. PhD thesis. Nijmegen: Radboud Universiteit Nijmegen. URL: http://hdl.handle.net/2066/160406.

Essegbey, James (1999). “Inherent complement verbs revisited: towards an understanding of argument structure in Ewe”. PhD thesis. Leiden University. URL: hdl.handle.net/21.11116/0000-0001-6D2F-2.

— (2019). *Tutrugbu (Nyangbo) Language and Culture*. Brill. DOI: 10.1163/9789004396999.

Fernando, Tim (2004). “A Finite-state Approach to Events in Natural Language Semantics”. In: *Journal of Logic and Computation* 14.1, pp. 79–92. DOI: 10.1093/logcom/14.1.79.

CLUL

Ginzburg, Jonathan (2012). *The interactive stance: Meaning for conversation*. Oxford: Oxford University Press. DOI: 10.1093/acprof:oso/9780199697922.001.0001.

Goldberg, Adele E. and Ray Jackendoff (2004). "The English Resultative as a Family of Constructions". In: *Language* 80.3, pp. 532–568. DOI: 10.1353/lan.2004.0129.

Harley, Matthew W. (2005). "A desciptive grammar of Tuwuli – A Kwa language of Ghana". PhD thesis. London: University of London.

Hiraiwa, Ken and Adams Bodomo (2008). "Object-sharing as symmetric sharing: predicate clefting and serial verbs in Dàgáárè". In: *Natural Language & Linguistic Theory* 26, pp. 795–832. DOI: 10.1007/s11049-008-9056-y.

Lau, Jonas (2021). "A digital reference grammar of Àbèsàbèsì". PhD thesis. Köln: Universität zu Köln.

Lefebvre, Claire and Anne-Marie Brousseau (2002). *Grammar of*

*Fonbge*. Mouton de Gruyter.


Lord, Carol (1974). “Causative constructions in Yorùbá”. In: *Studies in African Linguistics* 5.suppl, pp. 195–205.


— (1975). “Igbo compounds and the lexicon”. In: *Studies in African Linguistics* 6.1, pp. 23–48.


Madugu, Isaac S. George (1976). “Yorùbá adjectives have merged with verbs: or are they just emerging?” In: *Journal of West African Languages* 9.1-2, pp. 85–102.


Manfredi, Victor (1991). “Ágbọ̀ and Ẹ́hùgbò: Ìgbo linguistic concsciousness, its origins and limits”. PhD thesis. Cambridge, MA: Harvard University. URL: https://people.bu.edu/manfredi/dissertation.pdf.


— (2005). *Tense parameters and serial verbs*. Ms. University of Boston. URL: https://people.bu.edu/manfredi/svc.pdf.


— (2022). *The prosodic and semantic limits of serialization in the diachronic syntax of*

*Benue-Kwa*. Talk held at the 4th Symposium on West African Languages at L'Orientale di Napoli 22 September 2022. URL: https://people.bu.edu/manfredi/SyWalNapoli2022.pdf.


Müller, Stefan (2002). *Complex predicates: verbal complexes, resultative constructions and particle verbs in German*. Stanford, CA: CSLI.


Naumann, Ralf and Thomas Gamerschlag (2021). "Serial verb constructions and covert coordinations in Edo – an analysis in Type Logical Grammar". In: *Journal of Language Modelling* 8.2, pp. 337–413. DOI: 10.15398/jlm.v8i2.221.


Nwachukwu, Akụjụọobi Philip (1984). "Stative verbs in Igbo syntax". In: *The Journal of West African Languages* 14.2, pp. 81–101.


— (1987). "The argument structure of Igbo verbs". In: *Lexicon Project Working Papers* 18, pp. 1–142.


Ogie, Ota (2009). "Multi-verb constructions in Èdó". PhD thesis. Trondheim: Norwegian

University of Science and Technology.


Oshòdì, Bólúwají (2011). *A reference grammar of Arigidi*. Àkúrẹ́: Montem paperbacks.


Oyèlárăn, Ọlasope O. (1982). “On the scope of the serial verb construction in Yorùbá”. In: *Studies in African Lingustics* 13.2, pp. 109–146.


Rappaport Hovav, Malka and Beth Levin (1998). “Building verb meanings”. In: *The projections of arguments. Lexical and compositional factors*. Ed. by Miriam Butt and Wilhelm Geuder. Stanford, CA: Center for the Study of Language and Information, pp. 97–134.


Simpson, Jane (1983). “Resultatives”. In: *Papers in Lexical Functional Grammar*. Ed. by Lori S. Levin, Malka Rappaport, and Annie Zaenen. Indiana: Indiana University Linguistics Club, pp. 143–157. URL: http://hdl.handle.net/2123/140.


Stewart, Ọ̀sàmúyìmẹ̀n Thompson (2001). *The serial verb*

*construction parameter*. New York: Garland.


Van Valin, Robert D. and Randy Lapolla (1997). *Syntax: Form, meaning, and function*. Cambridge, MA: Cambridge University Press.


Wechsler, Stephen (2005). "Resultatives under the 'event-argument-homomorphism' model of telicity". In: *The syntax of aspect*. Ed. by Nomi Erteschick-Shir and Tova Rapoport. Oxford: Oxford University Press, pp. 255–273.


Wechsler, Stephen Mark (2001). "An Analysis of English Resultatives Under the Event-Argument Homomorphism Model of Telicity". In: *Proceedings of the 3rd Workshop on Text Structure, University of Texas, Austin, Oct. 13–15, 2000*.


Welmers, William Everett (1973). *African language structures*. Berkeley: University of California Press.


Zimmermann, Malte and Mary Amaechi (2025). "One, but not the same: on complex event-formation in

Igbo serial verb constructions". In: *Linguistics and Philosophy* 48.2, pp. 213–264. DOI: 10.1007/s10988-024-09412-w.